

Final Project Proposal: Model Predictive Control for Energy-Aware Racing Line Optimization of Electric Vehicles

Erwin Poussi

ERWINPI@STANFORD.EDU

Matthieu Hautsch

MATTHAUT@STANFORD.EDU

AA222/CS361, Stanford University

1. Problem

Finding the optimal racing line on a sinuous closed track is a classical problem in motorsport engineering. On straight sections, every driver can floor it and reach top speed, so there is little room for differentiation. But in the corners, performance differences emerge since the choice of racing line governs the maximum speed a car can carry through a turn, and small improvements in cornering translate into significant lap time gains (Xiong, 2010). Beyond the geometric question of which path to follow, the vehicle dynamics (tire grip limits, weight transfer, drift behavior) make this a rich nonlinear optimization problem.

We add an additional layer of complexity by considering electric vehicles (EVs). Unlike conventional race cars that can burn fuel freely, an EV must continuously manage its battery state of charge (*SOC*). Aggressive acceleration and high cornering speeds drain the battery faster through increased motor power demand and aerodynamic losses. Conversely, conservative driving preserves energy but sacrifices lap time. This creates a genuine dual-objective problem as we want to go fast *and* be energy-efficient. In the context of Formula E and the growing EV racing scene, this tradeoff is not just academic because it directly determines race strategy.

The objective of this project is to develop an optimization-based controller that computes racing trajectories for EVs on arbitrary closed circuits, jointly minimizing lap time and energy consumption. We consider a vehicle navigating a known 2D track and seek a control policy that determines steering, acceleration, and braking over time while respecting vehicle dynamics, tire grip constraints, track boundaries, and battery limits. This is inherently a constrained multi-objective optimization problem, combining trajectory planning (the spatial path) with speed profiling (how fast to drive along that path).

This problem fits squarely within the themes of AA222 since it involves dynamic systems modeling, nonlinear constrained optimization, and engineering design with competing objectives. It also connects to active research in eco-driving, autonomous racing, and predictive energy management (Sciarretta and Guzzella, 2007; Kochenderfer and Wheeler, 2019).

2. Proposed Approach

We formulate the problem as a finite-horizon *Model Predictive Control* (MPC) problem. At each timestep, the controller solves an optimization over a receding horizon, applies the first control action, then re-solves with updated state information.

Vehicle Model

We use a planar bicycle model, which captures both longitudinal and lateral dynamics while remaining tractable for real-time optimization. The state vector is $\mathbf{z} = [x, y, \psi, v_x, v_y, \omega, SOC]^T$, where (x, y) is position, ψ is heading angle, (v_x, v_y) are longitudinal and lateral veloci-

ties, ω is yaw rate, and $SOC \in [0, 1]$ is battery state of charge. The control inputs are $\mathbf{u} = [\delta, F_{\text{drive}}]^\top$, that is steering angle and net longitudinal force (positive for traction, negative for regenerative braking).

The longitudinal dynamics follow $m\dot{v}_x = F_{\text{drive}} - \frac{1}{2}\rho C_d A v_x^2 - C_r mg + mv_y \omega$, where $\rho C_d A$ captures aerodynamic drag and C_r is rolling resistance. Lateral and yaw dynamics are governed by tire forces through a simplified Pacejka-like model, with the friction circle constraint $F_x^2 + F_y^2 \leq (\mu mg)^2$ ensuring we do not exceed grip. The battery state evolves as $\dot{SOC} = -P_{\text{elec}}/(Q_{\text{batt}} \cdot V_{\text{nom}})$, where $P_{\text{elec}} = F_{\text{drive}} \cdot v_x/\eta$, with motor efficiency η (accounting for regen when $F_{\text{drive}} < 0$), battery capacity Q_{batt} , and nominal voltage V_{nom} .

Optimization Formulation

We minimize the weighted cost

$$J = \underbrace{w_t \cdot T_{\text{lap}}}_{\text{time}} + \underbrace{w_e \cdot \Delta SOC}_{\text{energy}} + \underbrace{w_u \int_0^T \|\dot{\mathbf{u}}\|^2 dt}_{\text{smoothness}}$$

subject to: speed bounds $\|\mathbf{v}\| \leq v_{\text{max}}$, actuation limits $|F_{\text{drive}}| \leq F_{\text{max}}$ and $|\delta| \leq \delta_{\text{max}}$, battery limits $SOC_{\text{min}} \leq SOC \leq 1$, tire grip $F_x^2 + F_y^2 \leq (\mu mg)^2$, and track boundary constraints. By varying w_t and w_e , we explore the Pareto front between lap time and energy consumption.

Experimental Setup

We plan to test the controller on several simulated closed circuits of varying complexity (from simple ovals to circuits with tight hairpins and chicanes), as shown in fig. 1.

Baselines. We will benchmark against: (1) constant-speed center-line following, (2) minimum-time aggressive driving (ignoring energy), (3) minimum-curvature path planning (Heilmeier et al., 2020), and (4) a heuristic energy-saving strategy.

Metrics. Performance will be evaluated on: lap time T_{lap} , energy consumed ΔSOC , ride smoothness $\int \|\dot{\mathbf{u}}\|^2 dt$, and solver runtime. We will also construct Pareto fronts between lap time and energy use.

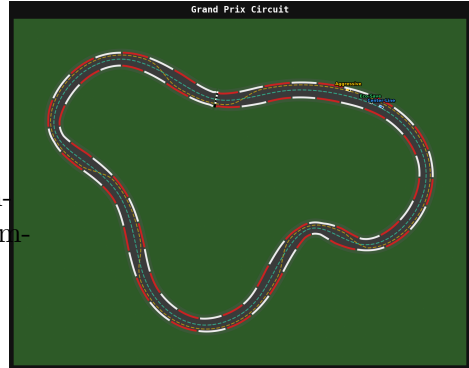


Figure 1: Simulated circuit with track boundaries and center-line.

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